

JOHN MARK OBURA

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EDUCATION

Oral Roberts University | Tulsa, OK

Expected May 2028

B.S. Electrical Engineering | Mathematics Minor | GPA: 4.00/4.00 | President's List

Additional Honors: Cambridge A Levels: 3 A*s; High School Valedictorian.

2022-2024

Courses: Network Analysis I & II; Electronics lecture & Lab; Engineering Computational Methods; Control Systems; Engineering Graphics; Differential Equations; Calculus III; Probability and Statistics; Digital Systems Design; Computer Networks and Communications; Microprocessors Design; Data Structures

TECHNICAL SKILLS

Circuits and Controls: Circuit analysis, rectification, filtering, voltage regulation, op-amp feedback, sensor signal conditioning, hysteresis, Arduino, embedded C/C++

Analysis and Tools: LTspice, MATLAB, Python, Excel, SolidWorks, Git/GitHub, R, C++, System Verilog

Laboratory: Oscilloscope, digital multimeter, function generator, DC power supply, breadboarding, soldering, circuit testing and troubleshooting

ENGINEERING PROJECTS

Adjustable 2-9 V Linear DC Power Supply | LTspice and Hardware Testing

- Designed, simulated and built a power supply using full-wave rectification, capacitive filtering, a Zener reference, op-amp feedback, and BJT regulation; selected components from circuit requirements.
- Automated load sweeps using .step and .meas; verified approximately 9 V output across 90.8-363.1 mA loads and 2.45 V peak-to-peak rectifier ripple below the 3.5 V limit.

Analog Temperature Controller | LTspice and Hardware Testing

- Designed, simulated, and built a thermistor-based controller with op-amp signal conditioning, Schmitt-trigger hysteresis, and BJT heater switching; characterized and modeled the sensor.
- Measured heater switching at approximately 86.7°F ON and 88.7°F OFF; compared hardware and simulation thresholds and evaluated measured op-amp saturation effects.

IR Maze-Solving Robot | Arduino and C/C++

- Integrated infrared sensors and servo control for real-time obstacle detection and autonomous navigation; one of three robots to complete the competition test course.

Bridge Dynamics Simulation | Python and NumPy

- Collaborated on an RK4 simulation of a mass-spring-damper bridge model under wind loading; plotted displacement to examine transient response, damping, and resonance.

32-bit RISC-V Processor | System Verilog

- Designed and functionally verified a processor, progressing from a single-cycle design to a five-stage pipeline with forwarding, hazard detection, and automated self-checking tests.

EXPERIENCE

Research Assistant | Oral Roberts University

Aug 2026 - Present

- Develop reproducible Python, PyTorch, and Qiskit simulation workflows for hybrid quantum-classical wireless-channel modeling; validate numerical outputs and analyze model behavior using probabilistic metrics.

IT Support Specialist | Oral Roberts University

Jan 2025 - Aug 2026

- Resolved 100+ hardware and software issues through SolarWinds Service Desk while applying CompTIA A+ and Network+ knowledge; collaborated with network administration on campus system updates and integrations.

LEADERSHIP

IEEE Student Branch Secretary | Oral Roberts University

- Organize hands-on circuit-design and soldering workshops to promote practical engineering skills.